

Match-the-Graph

Walk a position–time graph with the Go!Motion sensor · *Camina una gráfica de posición–tiempo*

Name _____ Date _____ Group _____

TODAY'S TARGET I can capture a position–time graph with a Vernier motion detector and explain what its slope tells me about velocity. <i>Puedo capturar una gráfica de posición–tiempo con un detector de movimiento Vernier y explicar qué me dice su pendiente sobre la velocidad.</i>	ASTEROID THREAD · 2026-XJ Radar gives us the asteroid's position over time — a graph. Today you learn to read that graph with your own body. Slope = velocity. <i>El radar nos da la posición del asteroide a lo largo del tiempo: una gráfica. Hoy aprendes a leer esa gráfica con tu propio cuerpo. Pendiente = velocidad.</i>
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LAB SETUP	
Driving question: Can you make your body draw a graph you were handed? · <i>¿Puedes hacer que tu cuerpo dibuje una gráfica?</i> Procedure 1. Open Graphical Analysis. Plug in the Go!Motion. Screen shows Position (m) vs Time (s). 2. Set the axes: time 0–10 s, position 0–4 m. 3. Read the target graph. Everyone predicts first (circle words). Then one person walks. 4. Press Collect. Walk. Sketch what the screen showed. 5. Talk: what would you change? Walk it once more. Rotate walker.	Equipment (per group of 3) • 1 Vernier Go!Motion + USB cable • 1 laptop with Graphical Analysis • 4 m clear lane; tape marks at 1, 2, 3 m • This sheet, one per person Safety: walk, don't run. Look where you are going, not at the screen — your partners read the screen. Stay inside the lane. <i>How this lab serves the year's question: if you can read a walk, you can read an asteroid.</i>

WORD BANK • BANCO DE PALABRAS

away • *se aleja*

steady • *constante*

slope • *pendiente*

toward • *se acerca*

faster • *más rápido*

flat • *plano*

stopped • *detenido*

slower • *más despacio*

steep • *empinado*

★ **The sensor is at position 0.** Walking **away** makes the line go **up**. Walking **toward** makes it go **down**.

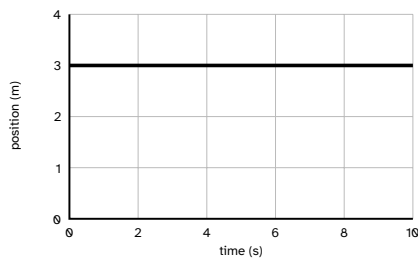
Part A • Predict, walk, sketch • Predice, camina, dibuja

For each graph: circle your prediction **before** anyone walks. Then sketch what the screen actually showed. If it did not match, write the one thing you would change.

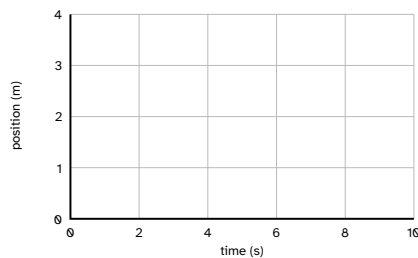
GRAPH 1

walker: _____

Target • Objetivo



Screen • Pantalla



Prediction: start [**far / close**], then [**stand still / walk away / walk toward**], [**slow / fast**].

Empieza [lejos / cerca], luego [se queda quieto / se aleja / se acerca], [despacio / rápido].

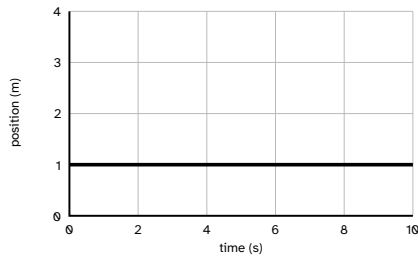
Matched? [**yes / almost / no**]

One thing we changed: _____

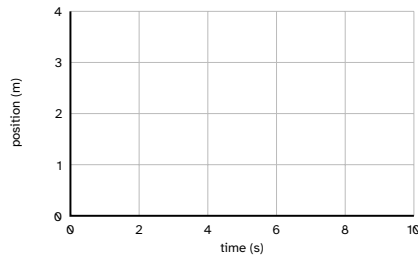
GRAPH 2

walker: _____

Target • Objetivo



Screen • Pantalla



Prediction: start [**far / close**], then [**stand still / walk away / walk toward**], [**slow / fast**].

Empieza [lejos / cerca], luego [se queda quieto / se aleja / se acerca], [despacio / rápido].

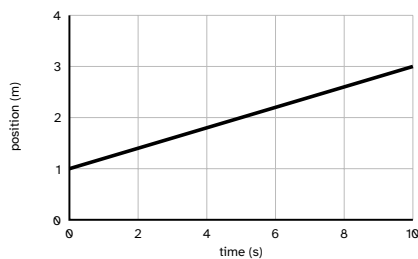
Matched? [**yes / almost / no**]

One thing we changed: _____

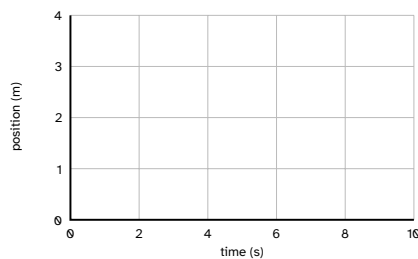
GRAPH 3

walker: _____

Target • Objetivo



Screen • Pantalla



Prediction: start [**far / close**], then [**stand still / walk away / walk toward**], [**slow / fast**].

Empieza [lejos / cerca], luego [se queda quieto / se aleja / se acerca], [despacio / rápido].

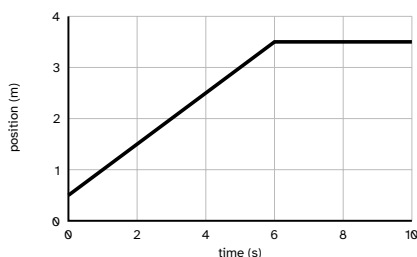
Matched? [**yes / almost / no**]

One thing we changed: _____

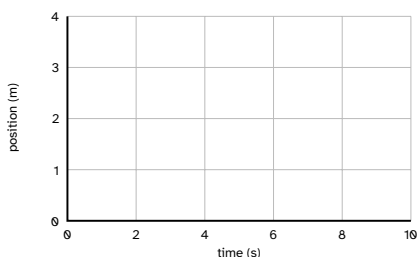
GRAPH 4

walker: _____

Target • Objetivo



Screen • Pantalla



Prediction: start [**far / close**], then [**stand still / walk away / walk toward**], [**slow / fast**].

Empieza [lejos / cerca], luego [se queda quieto / se aleja / se acerca], [despacio / rápido].

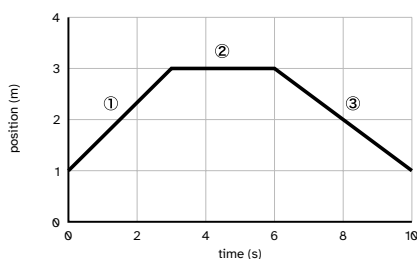
Matched? [**yes / almost / no**]

One thing we changed: _____

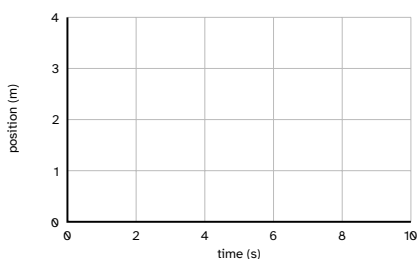
GRAPH 5

walker: _____

Target • Objetivo



Screen • Pantalla



Prediction — three parts: ① [**away / toward / still**] ② [**away / toward / still**] ③ [**away / toward / still**]

Tres partes: ① [se aleja / se acerca / quieto] ② ... ③ ...

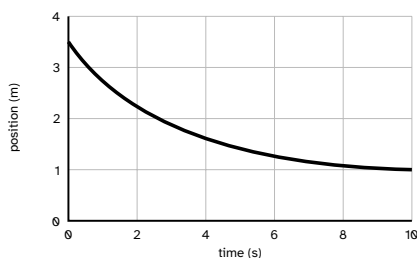
Matched? [**yes / almost / no**]

One thing we changed: _____

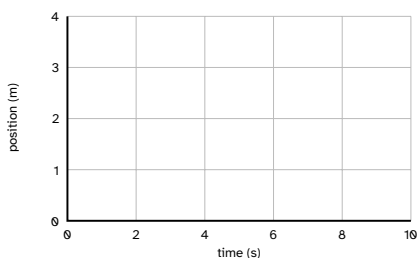
GRAPH 6

walker: _____

Target • Objetivo



Screen • Pantalla



Prediction: walk [**away / toward**], and get [**faster / slower / stay steady**].

Camina [alejándose / acercándose] y va [más rápido / más despacio / constante].

Matched? [**yes / almost / no**]

One thing we changed: _____

Part B • Slope = velocity • Pendiente = velocidad

Every straight part of a graph has a slope. The slope is the velocity. **Away from the sensor = positive (+). Toward the sensor = negative (-).**

WORKED EXAMPLE — GRAPH 3		
Graph 3 is one straight line. Pick two points on it: (0 s, 1 m) and (10 s, 3 m). Find the velocity.		
GIVEN	$t_1 = 0 \text{ s}, x_1 = 1 \text{ m} \cdot t_2 = 10 \text{ s}, x_2 = 3 \text{ m}$	<i>Read two points off the line.</i>
EQUATION	$v = \Delta x / \Delta t = (x_2 - x_1) / (t_2 - t_1)$	<i>Slope = rise over run.</i>
WORK	$v = (3 \text{ m} - 1 \text{ m}) / (10 \text{ s} - 0 \text{ s}) = 2 \text{ m} / 10 \text{ s}$	<i>Carry the units on every line.</i>
ANSWER	+0.2 m/s — This means the walker moved away from the sensor 0.2 m every second.	<i>Positive: away. Say it in words.</i>

YOUR TURN — GRAPH 4, SLOPED PART (0 s TO 6 s)	
GIVEN	$t_1 = 0 \text{ s}, x_1 = 0.5 \text{ m} \cdot t_2 = 6 \text{ s}, x_2 = \text{_____ m}$
EQUATION	$v = \Delta x / \Delta t$
WORK	
ANSWER	$v = \text{_____ m/s}$. This means the walker moved [away / toward] _____ m every second. Sign: [+ / -] because _____.

YOUR TURN — GRAPH 5, PART ① AND PART ③		
GIVEN	Part ① (0 s, 1 m) → (3 s, 3 m)	Part ③ (6 s, ___ m) → (10 s, ___ m)
EQUATION	$v = \Delta x / \Delta t$	
WORK		
ANSWER	$v_1 = \text{_____ m/s}$ sign [+ / -], so [away / toward]	$v_3 = \text{_____ m/s}$ sign [+ / -], so [away / toward]
Part ② is flat. Its slope is _____, so its velocity is _____ m/s. The walker was [stopped / moving steadily]. <i>La parte ② es plana. Su pendiente es ____, así que su velocidad es ___ m/s.</i>		

TONIGHT, ONLINE • ESTA NOCHE, EN LA APP

1 • Same idea, new object. You will type 2026-XJ's position–time data and watch the graph build. Ask: is it flat, rising, or falling? Steep or gentle?

2 • Same math. Radar logs 18,000 km at 30 min and 36,000 km at 60 min. Find the slope, like Part B — but the units will be **km/h**.

3 • A graph is a claim. Bring one graph from today. You will say what it claims and what walk would prove it wrong.

Part C • Reverse it • Al revés

Now the screen goes second. A partner writes a walk in words. You sketch the graph it should make **before** anyone moves. Then the partner walks it and you compare.

PARTNER'S WALK • WRITTEN BY: _____

Start at _____ m. For _____ s, [**stand still / walk away / walk toward**] [**slow / fast**].
Then for _____ s, [**stand still / walk away / walk toward**] [**slow / fast**].
Then for _____ s, [**stand still / walk away / walk toward**] [**slow / fast**].
*Empieza en ____ m. Durante ____ s, [**quieto / se aleja / se acerca**] [**despacio / rápido**]...*

My sketch — BEFORE the walk • Mi dibujo, ANTES

A blank coordinate grid for a position-time graph. The vertical axis is labeled 'position (m)' and has tick marks from 0 to 4. The horizontal axis is labeled 'time (s)' and has tick marks from 0 to 10 in increments of 2.

What the screen showed • Lo que mostró la pantalla

A blank coordinate grid for a position-time graph, identical to the one on the left. The vertical axis is labeled 'position (m)' and has tick marks from 0 to 4. The horizontal axis is labeled 'time (s)' and has tick marks from 0 to 10 in increments of 2.

My sketch and the screen [**matched / did not match**] because _____.

*Mi dibujo y la pantalla [**coincidieron / no coincidieron**] porque _____.*

EXIT TICKET — DAY 3

A position–time graph is **flat** from 4 s to 7 s. A classmate says, "The walker moved at a constant speed during those 3 seconds." Are they right? Explain using the word **slope**. Write, sketch, or both. • *¿Tiene razón? Explica usando la palabra pendiente.*

CPA Physics • Unit 1 • Day 3 • Investigation 1.1

Match-the-Graph

Today's Target — Rate yourself:

1 = Not Yet (I can't do this without help) · **2 = Almost** (I can do this with some help or some errors) · **3 = Got It** (I can do this independently and explain it to someone else)

Target	1	2	3	Help?
I can capture a position–time graph with a Vernier motion detector and explain what its slope tells me about velocity.	☐	☐	☐	☐